

1. Ans:- str1 = "Hello World"
Rev = str1[::-1]

2. Ans:- str = 'A Quick Brown Fox Jump Over the Lazy Dog'
Str1=x.split()
Print(len(str1))

3. Ans:- str1=carrot

4. Difference Between List and set are:-

lists	sets
List are ordered ✓	Sets are unordered ✓
List allows duplicate ✓	It doesnot allow duplicate ✓
It is slower ✓	It is faster ✓
It is created by using [] ✓	It is created by using { } ✓

5. Ans:- No sets doesnot contains Lists because sets elements are Immutable .

6. Ans:- Sets is faster because it uses hash table which is faster than other data structure.

7. Remove() also deletes the elements but it throws error if elements not exist while
discard() also removes the elements but it doesnot throws error which is safest method.

8. Ans:- There are 3 methods to insert elements in the list.they are append(), extend(),
insert() ✓ ✓ ✓

9. Ans:- There are 5 methods to delete elements in the list.they are remove(), pop(n),
pop(), clear(), delete() ✓ ✓ ✓

10. There are 6 operators in python. They are :-

- ✓ 1. Assignment Operator:- It contains (=) equals, it assigns the value to the variable
- ✓ 2. Arithmetic Operator:- it contains +, -, *, /, //, **, %, it is used to perform arithmetic operations
- ✓ 3. Logical Operator:- it contains and, or, not operator, it is used to perform Boolean operation. It returns true or false
- ✓ 4. Relational Operator:- It contains >, <, >=, <=, !=, ==, it is used to compare between the two operands. It returns true or false
- ✓ 5. Identity Operator :- It contains is, is not, . It returns true or false
- ✓ 6. Membership operator :- It contains in, not in, . It returns true or false

11. Ans:- It helps to perform operation based on their precedence,
Precedence of operator as :- (), *, /, //, **, +, -, -

12. Ans :- sort the list in python using slicing method `list1=[5,11,14,7,8,3,2,5,9]` `asc=`
`list1.sort()`
`desc = asc[::-1]`

13. Dictionary is a data type in python which is collection of key value pairs. Each key in a dictionary must be unique. It is created using { }. It is mutable. Eg `student = {'name': 'ankesh', 'roll': 4}`

14. There are 3 methods to create a dictionary.

1. Using { } there is key and value pair
2. Dict ()
3. { } empty dictionary

15. There are 2 methods to access value. they are

1. By using key eg. `Student['name']`
2. By using get method eg.
`Student.get('name')`

16. There are 4 methods to delete elements in dictionary.

- a. `Pop(n)`
- b. `Pop()`

popitem()
pop(Student['age'])

- c. Clear()
- d. Delete()

17. Ans :- we can perform operation in sets in 4 methods

- a. Union (|)
- b. Intersection(&)
- c. Difference (-)
- d. Symmetric difference (^)

18. Packing means storing multiple value in tuple like combination of string,int,float,Boolean . eg . tup1 = (1,2.3,'ram')

19. Unpacking means extracting tuple value into variables Eg . lang =('c','c++','python')
lan1,lan2,lan3 =lang

20. Difference between sets and dictionary are as follows :-

Sets	Dictionary
Duplicates not allowed	Duplicate key is not allowed but duplicate value allowed
Only element is present	Key value pair
Element must be unique	Key must be unique